

Computer Vision Bootcamp

A collection of professional Computer Vision certificates covering Transformers, Computer vision application, deep learning for vision, and real-world applications.

- **Deep Learning for Computer Vision**
- **Computer Vision Application**
- **Transformers in computer vision**
- **Reinforcement learning**
- **Deployment of ML Models**

Machine Learning Certificate

What truly made this course special was how deeply it explained the low-level implementation of machine learning algorithms — learning to code them from scratch.

We covered:

- **Part 1: Machine Learning**
- **Linear & Polynomial Regression**
- **Regularization (Ridge, Lasso)**
- **Binary & Multiclass Classification**
- **Neural Networks**
- **Unsupervised Learning (Autoencoder, K-Means, GMM)**
- **KNN & Probabilistic Modeling**

We also learned key aspects of any ML project:

- **Model Selection, Data Collection, Feature Engineering**

- **Data Augmentation (Synthetic Data)**
- **Handling Data Leakage**
- **Building End-to-End ML Projects**

Part 2: Introduction to Deep Learning

- **PyTorch Framework**
- **CNN, RNN, and LSTM Networks**

Data Structures & Algorithms Certificate

I recently completed the Data Structures & Algorithms course with CSkilled Academy, and it was an incredible experience!

- **Deep Dive into Data Structures:** Implemented vectors, linked lists, stacks, queues, trees, heaps, tries, and hash tables.
- **Algorithm Mastery:** Covered sorting, binary search, graph algorithms, dynamic programming, backtracking, and divide & conquer.
- **Problem Solving:** Solved numerous coding challenges of varying difficulty.
- **Hands-on Practice:** Reinforced concepts with real-world algorithmic challenges and deep implementation focus.

Python & OOP Certificate

My journey with Python and OOP at CSkilled Academy was an incredible experience where I gained a solid foundation in both coding and software design.

- **Solid Foundation:** Mastered Python fundamentals and object-oriented programming concepts.
- **Problem Solving:** Solved challenges of increasing difficulty to sharpen my coding skills.
- **Hands-on Projects:** Built 5 projects with increasing complexity to apply practical skills.
- **Advanced Topics:** Explored multithreading, networking, regex, bitwise operations, and decorators.

Computer Vision Certificates

I completed a structured learning path on DataCamp, progressing from the fundamentals to advanced deep learning problems.

Topics Covered:

- **Introduction to Deep Learning with PyTorch:** Neural network architectures, hyperparameters, model training, and evaluation.
- **Intermediate Deep Learning:** Building robust CNN and RNN models, handling images and sequences.
- **Deep Learning for Images:** Image classification, object recognition, segmentation, and GAN-based image generation.

HTML, CSS & JavaScript Certificates

- Completed the JavaScript & ES6 course on MaharaTech - ITIMooca.
- Finished the 30-Day JavaScript Challenge on LeetCode, covering closures, array & function transformations, promises, timing, JSON, and classes.
- Completed the 22-Week JavaScript Challenge with Osama Elzero, focusing on core JS fundamentals, plus hands-on HTML & CSS practice.

SQL Database Certificates

I completed the “Database Fundamentals” course, where I learned DBMS principles, ERD creation, relational database design, and SQL query writing.

Then, I took “Transact SQL Queries using SQL Server” with Engineer Rami Mohamed — a comprehensive course covering everything from basics to advanced SQL concepts.

- Creating and modifying databases using T-SQL
- Filtering, sorting, and combining data using SQL operations
- Using LIKE, subqueries, set operations, and EERD modeling
- Database integrity, constraints, and indexing strategies
- Backups, snapshots, and high-availability systems (mirroring, log shipping)
- Stored procedures, triggers, and CLR-based functions
- Automation with SQL Server Management Objects (SMO)
- Dynamic report design using SSRS
- Data warehousing concepts (OLAP vs. OLTP, dimensional modeling)

Each chapter included practical, hands-on assignments to reinforce the lessons and develop real-world database expertise.